Optimal rates for the kernel ridge regression

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1 Introduction

Standard learning framework: $X, Y \in \mathcal{X} \times \mathcal{Y} \sim \rho$, $\mathcal{X}$ is a Polish space with $\mathcal{Y}$ a separable Hilbert space.
Training data $\{X_i, Y_i, i = 1, \ldots, n\}$ are IID drawn from ρ.

Assumption 1 (RKHS)**.** *$\mathcal{H}$ is a separable Hilbert space of functions $f : X \to Y$. In particular,*

1. *there exists a Hilber-Schmidt operator $K_x : Y \to \mathcal{H}$ such that*

$$K_x^* f = f(x), \forall f \in \mathcal{H},$$

where $K_x^* : \mathcal{H} \to Y$ is the adjoint operator of K_x ($\langle f, K_x y\rangle_{\mathcal{H}} = \langle K_x^* f, y\rangle_{\mathcal{Y}}$).

2. *for any orthonormal basis $\{e_i\}_i$ of $\mathcal{Y}$, $\mathrm{tr}(K_x^* K_x) = \sum_i \langle K_x e_i, K_x e_i\rangle_{\mathcal{H}} \leq \kappa < \infty$.*

3. *$(x, t) \to \langle K_x v, k_t w\rangle$ is measurable for any $v, w \in \mathcal{Y}$.*

Let $T_x = K_x K_x^* : \mathcal{H} \to \mathcal{H}$ and $T = \int_{\mathcal{X}} T_x d\rho_X(x)$. Note T is a positive self-adjoint trace-class operator with $\|T\|_{\mathcal{L}(\mathcal{H})} \leq \kappa$, hence has spectrum $t_1 \geq t_2 \geq ... \geq t_N > 0$ (N can be ∞).

Remark 1. *The kernel is $k(x, t) = K_x^* K_t : \mathcal{X} \times \mathcal{X} \to \mathcal{L}(Y)$, $\mathcal{L}(Y)$ is the Hilbert space of all bounded linear operator on Y.*

Remark 2. *When $\mathcal{Y} = \mathbb{R}$, K_x is reloaded by $K_x 1$, and $k(x, t) = \langle K_x, K_t\rangle_{\mathcal{H}}$, thus recover the classical RKHS setting. Furthermore, $(Tf)(x) = \int_{\mathcal{X}} k(x, t) f(t) d\rho_X(t)$, i.e., T is an integral operator.*

The risk is

$$R(f) = \mathbb{E}_\rho \|f - Y\|^2.$$

Try to fit a function such that

$$\widehat{f}_\lambda = \arg\min_{f \in \mathcal{H}} n^{-1} \sum_{i=1}^{n} \|Y_i - f(X_i)\|^2 + \lambda \|f\|_{\mathcal{H}}^2.$$

To study the convergence rate of $\widehat{f}_\lambda$, we require the data distribution to fall in families $\mathcal{P}(b, c)$ satisfying following conditions with $1 < b \leq +\infty, 1 \leq c \leq 2$.

Assumption 2 (Function class)**.** *For all distributions in $\mathcal{P}(b,c)$, we have*

1. $\mathbb{E}\|Y\|^2 < \infty$
2. *There exists $f_{\mathcal{H}} \in \mathcal{H}$ such that*
$$f_{\mathcal{H}} = \arg\min_{f \in \mathcal{H}} R(f).$$
3. *There exist constants Σ, M such that*
$$\int_{\mathcal{Y}} [\exp(-\|y - f_{\mathcal{H}}(x)\|/M) - \|y - f_{\mathcal{H}}(x)\|/M - 1] d\rho(y|x) \leq \frac{\Sigma^2}{2M^2} \tag{1}$$
for ρ_X-almost all x.
4. *There exist $g \in \mathcal{H}$ and constant B such that*
$$f_{\mathcal{H}} = T^{(c-1)/2} g, \|g\|_{\mathcal{H}}^2 \leq B. \tag{2}$$
5. *If $b < \infty$, then $N = \infty$, and there exist constants α, β such that the eigenvalues of T satisfying*
$$\alpha \leq n^b t_n \leq \beta, \forall n \geq 1, \tag{3}$$
otherwise when $b = \infty$, $N \leq \beta < \infty$.

Remark 3. *Eq. (1) is a requirement on the noise distribution. Eq. (3) states that the eigenvalues of T has exponential decay. $b = \infty$ means the effective dimension of $\mathcal{H}$ is finite. c is a measure of the complexity of $f_{\mathcal{H}}$. If $\mathcal{H}$ is a Sobolev space, c is related with the smoothness of $f_{\mathcal{H}}$. Eq. (2) holds for any $1 < c \leq 2$ if $b = \infty$ since T is invertible when $\mathcal{H}$ is finite dimensional.*

2 Upper and lower bounds

Theorem 1 (Caponnetto and De Vito (2007))**.** *Given $1 < b \leq \infty$ and $1 \leq c \leq 2$, let*
$$\lambda_n = \begin{cases} n^{-b/(bc+1)}, & b < \infty, c > 1, \\ (\log n/n)^{b/(b+1)}, & b < \infty, c = 1, \\ n^{-1/2}, & b = \infty, \end{cases}$$
and
$$a_n = \begin{cases} n^{-bc/(bc+1)}, & b < \infty, c > 1, \\ (\log n/n)^{b/(b+1)}, & b < \infty, c = 1, \\ n^{-1}, & b = \infty, \end{cases}$$
then
$$\lim_{\tau \to \infty} \limsup_{n \to \infty} \sup_{\rho \in \mathcal{P}(b,c)} pr(R(\widehat{f}_{\lambda_n}) - R(f_{\mathcal{H}}) > \tau a_n) = 0.$$

Remark 4. *This theorem shows that a_n is an upper bound for the convergence rate. The minimax lower bound matches this bound for $b < \infty$, hence is optimal.*

3 Proof sketch

Proposition 1. *The following facts hold under Assumptions 1 and 2.*

1. *For all* $f \in \mathcal{H}$, $R(f) - R(f_{\mathcal{H}}) = \|f - f_{\mathcal{H}}\|_Y^2 = \|\sqrt{T}(f - f_{\mathcal{H}})\|_{\mathcal{H}}^2$.

   Proof.

$$\begin{aligned}\|\sqrt{T}f\|_{\mathcal{H}}^2 &= \langle \int_{\mathcal{X}} K_x K_x^* f d\rho_X(x), f\rangle_{\mathcal{H}} \\ &= \int_{\mathcal{X}} \langle K_x K_x^* f, f\rangle_{\mathcal{H}} d\rho_X(x) \\ &= \int_{\mathcal{X}} \langle K_x^* f, K_x^* f\rangle_Y d\rho_X(x) = \int_{\mathcal{X}} f^2(x) d\rho_X(x) = \|f\|_Y^2.\end{aligned}$$

   □

2. *For any* $\lambda > 0$,

$$\begin{aligned}f_\lambda &= (T + \lambda)^{-1} g = (T + \lambda)^{-1} T f_{\mathcal{H}}. \\ \widehat{f}_\lambda &= (T_{\boldsymbol{x}} + \lambda)^{-1} g_{\boldsymbol{z}},\end{aligned}$$

   where $T_{\boldsymbol{x}} = n^{-1}\sum_{i=1}^n T_{x_i}$, $g_{\boldsymbol{z}} = n^{-1}\sum_{i=1}^n K_{x_i} y_i$.

   Proof. Recall

$$f_\lambda = \arg\min_{f \in \mathcal{H}} R(f) + \lambda\|f\|_{\mathcal{H}}^2 = \arg\min_{f \in \mathcal{H}} \|\sqrt{T}(f - f_{\mathcal{H}})\|_{\mathcal{H}}^2 + \lambda\|f\|_{\mathcal{H}}^2. \tag{4}$$

$$\widehat{f}_\lambda = \arg\min_{f \in \mathcal{H}} n^{-1}\sum_{i=1}^n \|Y_i - f(X_i)\|^2 + \lambda\|f\|_{\mathcal{H}}^2. \tag{5}$$

   So taking Fréchet derivative to Eq. (4), f_λ satisfies

$$T(f_\lambda - f_{\mathcal{H}}) + \lambda f_\lambda = 0.$$

   Similarly, the Fréchet derivative of Eq. (5) is

$$\begin{aligned}\frac{\partial}{\partial f}\left[n^{-1}\sum_{i=1}^n \|y_i - K_{x_i}^* f\|^2 + \lambda\|f\|_{\mathcal{H}}^2\right] &= 2n^{-1}\sum_{i=1}^n K_{x_i}(K_{x_i}^* f - y_i) + 2\lambda f \\ &= 2(T_{\boldsymbol{x}} f + \lambda f - g_{\boldsymbol{z}}).\end{aligned}$$

   □

Idea: decompose the error as the bias and variance term.

$$\widehat{f}_\lambda - f_{\mathcal{H}} = \widehat{f}_\lambda - f_\lambda + f_\lambda - f_{\mathcal{H}}.$$

The bias term $f_\lambda - f_{\mathcal{H}}$ is known to converge to zero as $\lambda \to 0$ (recall prop above).

The variance term

$$\begin{aligned}
\widehat{f}_\lambda - f_\lambda &= \left((T_{\mathbf{x}} + \lambda)^{-1} g_{\mathbf{z}}\right) - \left((T+\lambda)^{-1} g\right) \\
&= (T_{\mathbf{x}} + \lambda)^{-1} \left\{(g_{\mathbf{z}} - g) + (T - T_{\mathbf{x}})(T+\lambda)^{-1} g\right\} \\
&= (T_{\mathbf{x}} + \lambda)^{-1} \{(g_{\mathbf{z}} - T_{\mathbf{x}} f_{\mathcal{H}} + T_{\mathbf{x}} f_{\mathcal{H}} - T f_{\mathcal{H}}) + (T - T_{\mathbf{x}}) f_\lambda\} \\
&= (T_{\mathbf{x}} + \lambda)^{-1} (g_{\mathbf{z}} - T_{\mathbf{x}} f_{\mathcal{H}}) + (T_{\mathbf{x}} + \lambda)^{-1} (T - T_{\mathbf{x}}) \left(f^\lambda - f_{\mathcal{H}}\right).
\end{aligned}$$

Note that $T_{\boldsymbol{x}}$ will converge to T as $n \to \infty$, and $g_{\boldsymbol{z}} - T_{\boldsymbol{x}} f_{\mathcal{H}}$ is controlled by the concentration inequality ($\mathbb{E} g_{\boldsymbol{z}} = T_{\boldsymbol{x}} f_{\mathcal{H}}$).
The error bound of $R(\widehat{f}_\lambda) - R(f_{\mathcal{H}})$ is dominated by three factors:

1. The residual

$$\mathcal{A}(\lambda) = \|f_\lambda - f_{\mathcal{H}}\|_Y^2.$$

2. The reconstruction error

$$\mathcal{B}(\lambda) = \|f_\lambda - f_{\mathcal{H}}\|_{\mathcal{H}}^2.$$

3. The effective dimension

$$\mathcal{N}(\lambda) = \mathrm{tr}[(T+\lambda)^{-1} T].$$

References

Andrea Caponnetto and Ernesto De Vito. Optimal rates for the regularized least-squares algorithm. *Foundations of Computational Mathematics*, 7(3):331–368, 2007.